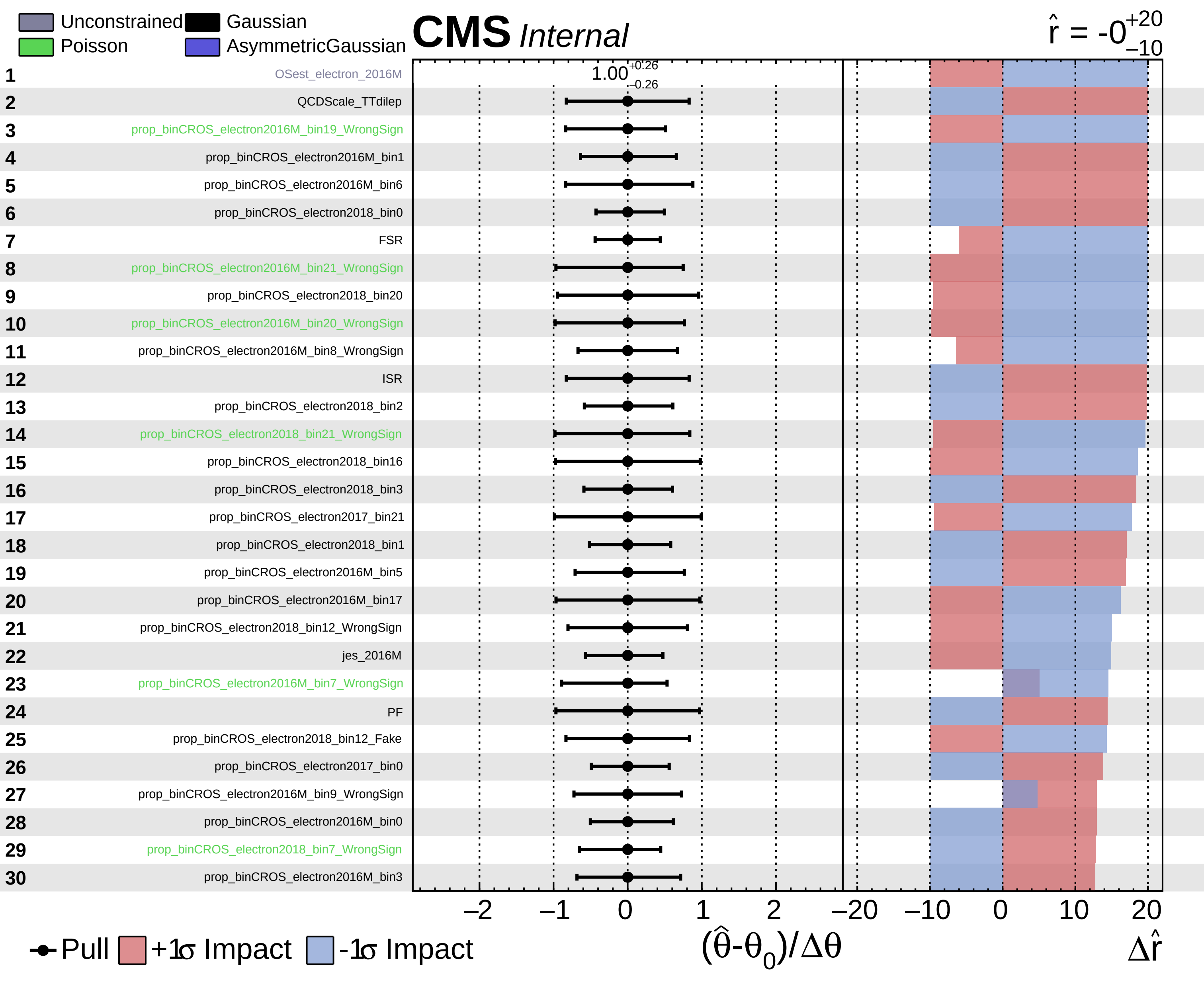
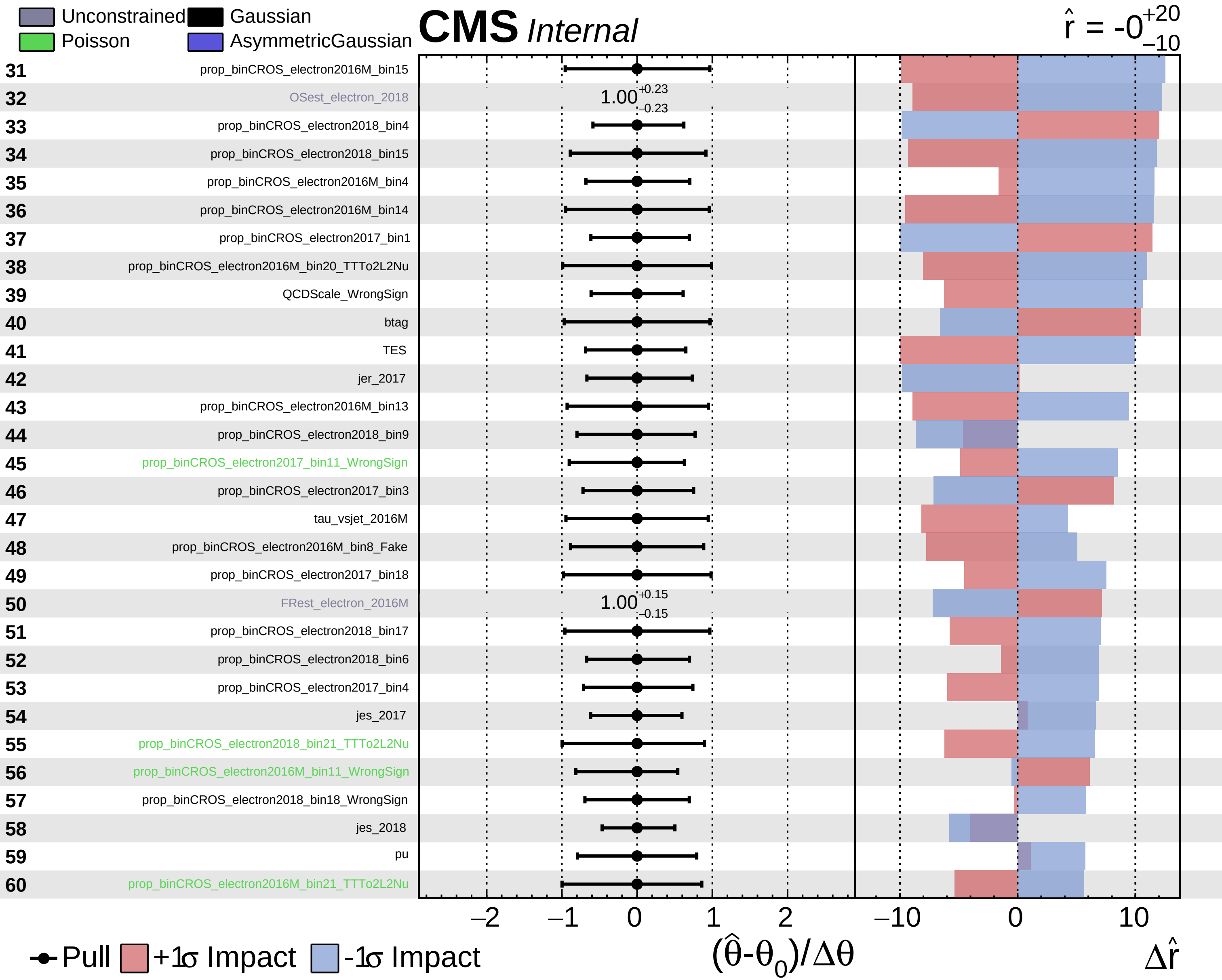


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0^{+20}_{-10}$
$1.00^{+0.26}_{-0.26}$
1 OSest_electron_2016M
2 QCDScale_TTdilep
3 prop_binCROS_electron2016M_bin19_WrongSign
4 prop_binCROS_electron2016M_bin1
5 prop_binCROS_electron2016M_bin6
6 prop_binCROS_electron2018_bin0
7 FSR
8 prop_binCROS_electron2016M_bin21_WrongSign
9 prop_binCROS_electron2018_bin20
10 prop_binCROS_electron2016M_bin20_WrongSign
11 prop_binCROS_electron2016M_bin8_WrongSign
12 ISR
13 prop_binCROS_electron2018_bin2
14 prop_binCROS_electron2018_bin21_WrongSign
15 prop_binCROS_electron2018_bin16
16 prop_binCROS_electron2018_bin3
17 prop_binCROS_electron2017_bin21
18 prop_binCROS_electron2018_bin1
19 prop_binCROS_electron2016M_bin5
20 prop_binCROS_electron2016M_bin17
21 prop_binCROS_electron2018_bin12_WrongSign
22 jes_2016M
23 prop_binCROS_electron2016M_bin7_WrongSign
24 PF
25 prop_binCROS_electron2018_bin12_Fake
26 prop_binCROS_electron2017_bin0
27 prop_binCROS_electron2016M_bin9_WrongSign
28 prop_binCROS_electron2016M_bin0
29 prop_binCROS_electron2018_bin7_WrongSign
30 prop_binCROS_electron2016M_bin3
−2 −1 0 1 2
−20 −10 0 10 20
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
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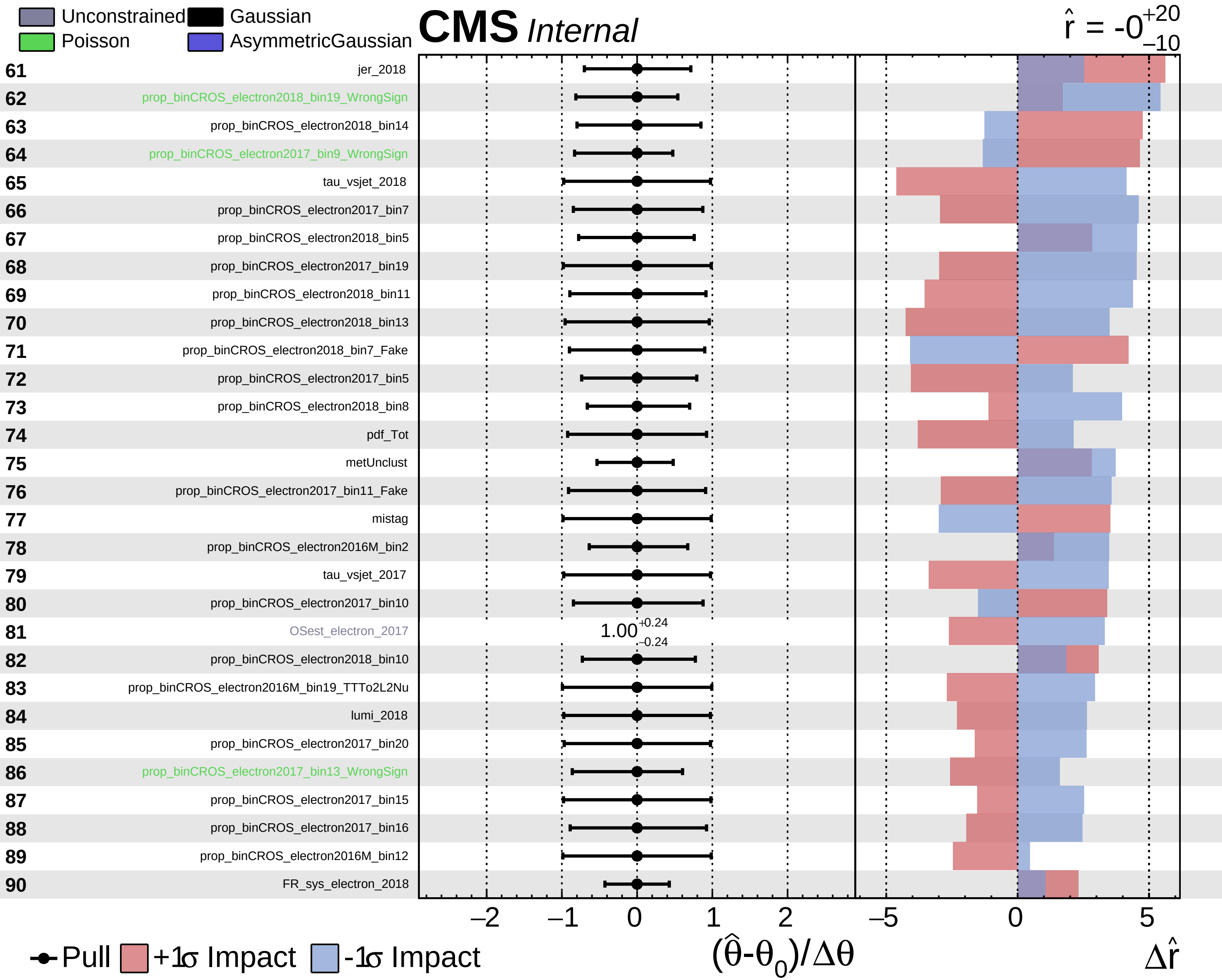
31 prop_binCROS_electron2016M_bin15
32 OSest_electron_2018
$1.00^{+0.23}_{-0.23}$
33 prop_binCROS_electron2018_bin4
34 prop_binCROS_electron2018_bin15
35 prop_binCROS_electron2016M_bin4
36 prop_binCROS_electron2016M_bin14
37 prop_binCROS_electron2017_bin1
38 prop_binCROS_electron2016M_bin20_TTTo2L2Nu
39 QCDScale_WrongSign
40 btag
41 TES
42 jer_2017
43 prop_binCROS_electron2016M_bin13
44 prop_binCROS_electron2018_bin9
45 prop_binCROS_electron2017_bin11_WrongSign
46 prop_binCROS_electron2017_bin3
47 tau_vsjet_2016M
48 prop_binCROS_electron2016M_bin8_Fake
49 prop_binCROS_electron2017_bin18
50 FRest_electron_2016M
$1.00^{+0.15}_{-0.15}$
51 prop_binCROS_electron2018_bin17
52 prop_binCROS_electron2018_bin6
53 prop_binCROS_electron2017_bin4
54 jes_2017
55 prop_binCROS_electron2018_bin21_TTTo2L2Nu
56 prop_binCROS_electron2016M_bin11_WrongSign
57 prop_binCROS_electron2018_bin18_WrongSign
58 jes_2018
59 pu
60 prop_binCROS_electron2016M_bin21_TTTo2L2Nu
−2
−1
0
1
2
−10
0
10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



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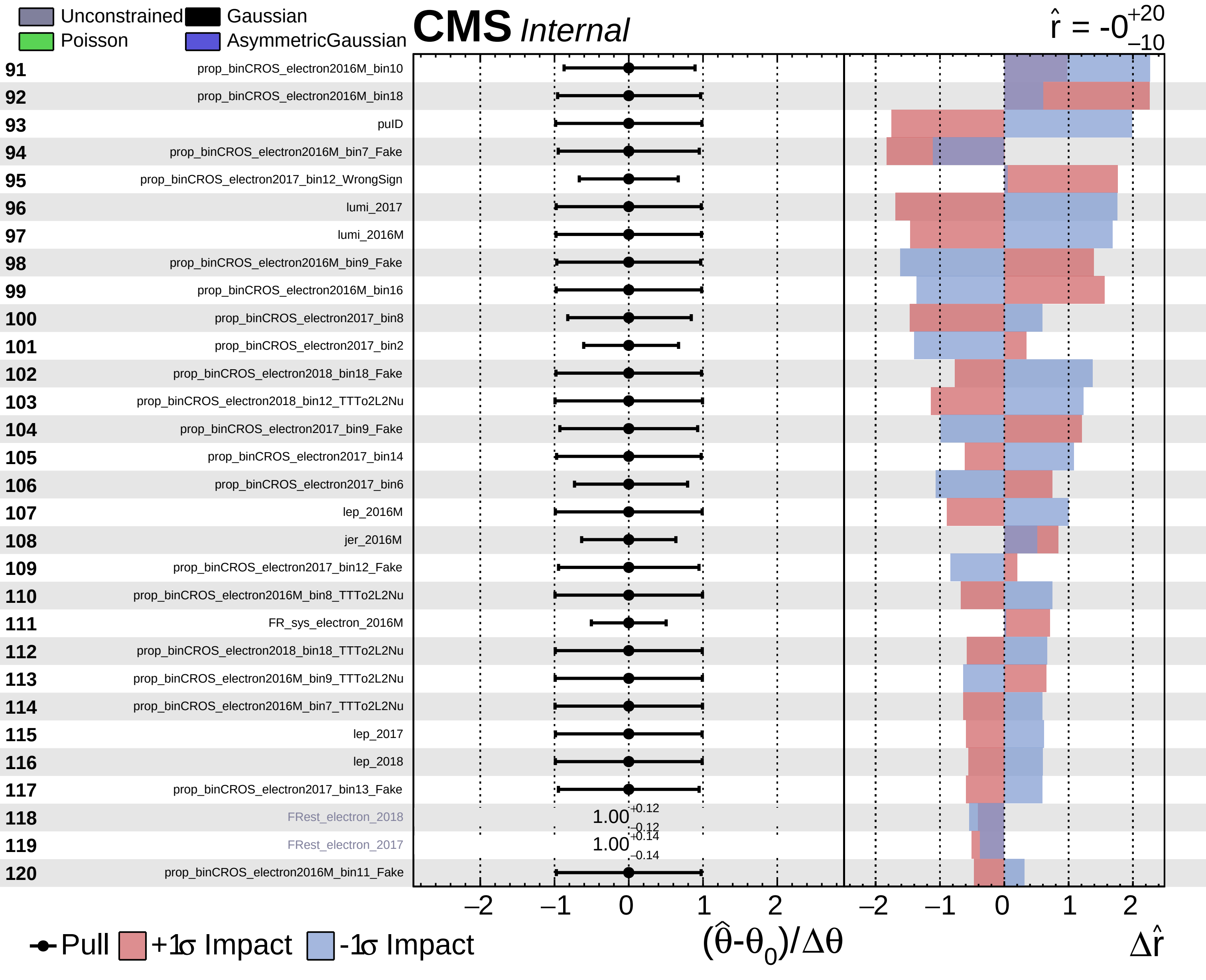
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0^{+20}_{-10}$
61 jer_2018
62 prop_binCROS_electron2018_bin19_WrongSign
63 prop_binCROS_electron2018_bin14
64 prop_binCROS_electron2017_bin9_WrongSign
65 tau_vsjet_2018
66 prop_binCROS_electron2017_bin7
67 prop_binCROS_electron2018_bin5
68 prop_binCROS_electron2017_bin19
69 prop_binCROS_electron2018_bin11
70 prop_binCROS_electron2018_bin13
71 prop_binCROS_electron2018_bin7_Fake
72 prop_binCROS_electron2017_bin5
73 prop_binCROS_electron2018_bin8
74 pdf_Tot
75 metUnclust
76 prop_binCROS_electron2017_bin11_Fake
77 mistag
78 prop_binCROS_electron2016M_bin2
79 tau_vsjet_2017
80 prop_binCROS_electron2017_bin10
81 OSest_electron_2017
$1.00^{+0.24}_{-0.24}$
82 prop_binCROS_electron2018_bin10
83 prop_binCROS_electron2016M_bin19_TTTo2L2Nu
84 lumi_2018
85 prop_binCROS_electron2017_bin20
86 prop_binCROS_electron2017_bin13_WrongSign
87 prop_binCROS_electron2017_bin15
88 prop_binCROS_electron2017_bin16
89 prop_binCROS_electron2016M_bin12
90 FR_sys_electron_2018
−2 −1 0 1 2
−5 0 5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0 +20 −10
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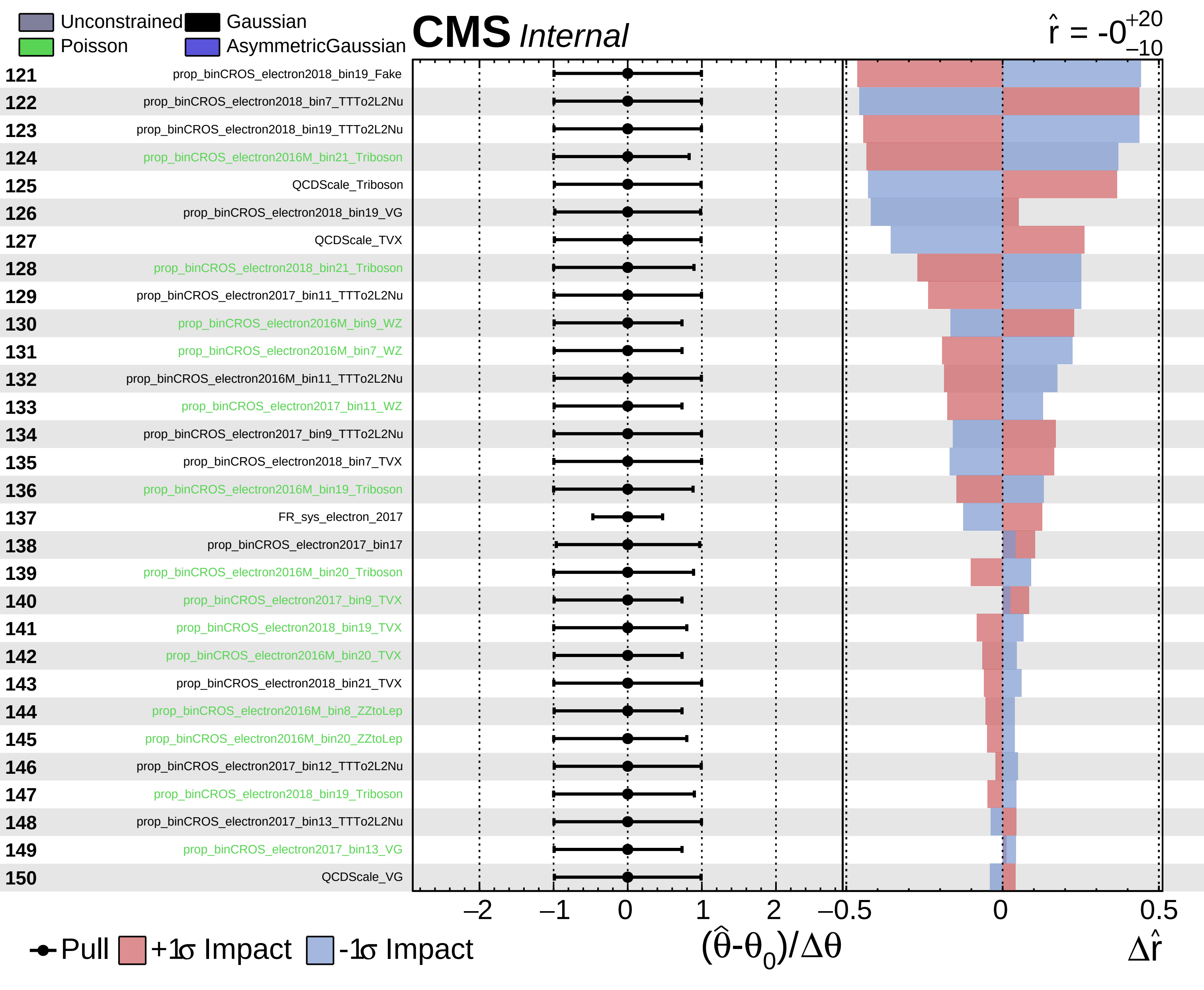
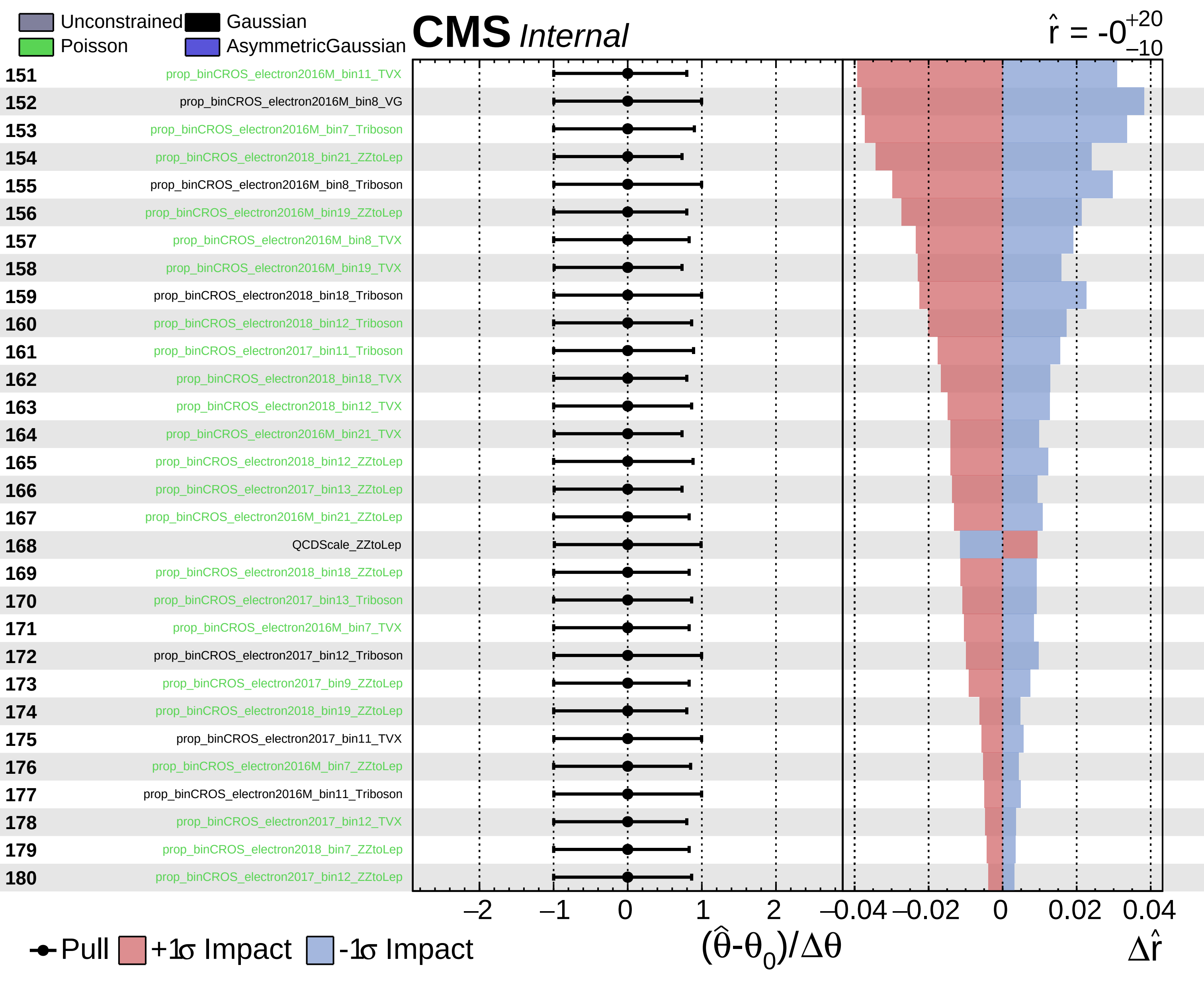
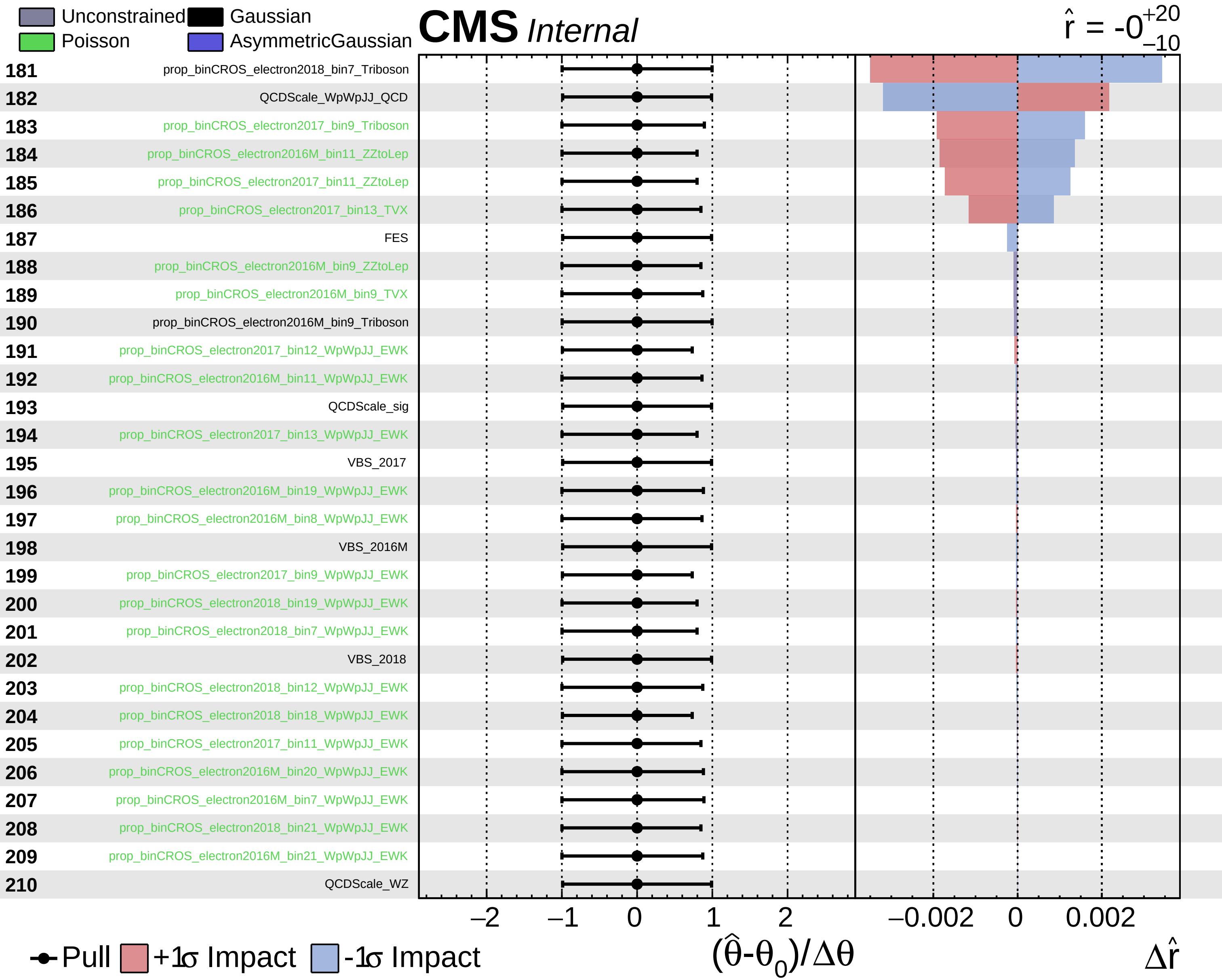
92 prop_binCROS_electron2016M_bin18
93 puID
94 prop_binCROS_electron2016M_bin7_Fake
95 prop_binCROS_electron2017_bin12_WrongSign
96 lumi_2017
97 lumi_2016M
98 prop_binCROS_electron2016M_bin9_Fake
99 prop_binCROS_electron2016M_bin16
100 prop_binCROS_electron2017_bin8
101 prop_binCROS_electron2017_bin2
102 prop_binCROS_electron2018_bin18_Fake
103 prop_binCROS_electron2018_bin12_TTTo2L2Nu
104 prop_binCROS_electron2017_bin9_Fake
105 prop_binCROS_electron2017_bin14
106 prop_binCROS_electron2017_bin6
107 lep_2016M
108 jer_2016M
109 prop_binCROS_electron2017_bin12_Fake
110 prop_binCROS_electron2016M_bin8_TTTo2L2Nu
111 FR_sys_electron_2016M
112 prop_binCROS_electron2018_bin18_TTTo2L2Nu
113 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
114 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
115 lep_2017
116 lep_2018
117 prop_binCROS_electron2017_bin13_Fake
118 FRest_electron_2018
1.00 +0.12 −0.12
119 FRest_electron_2017
1.00 +0.14 −0.14
120 prop_binCROS_electron2016M_bin11_Fake
−2 −1 0 1 2
−2 −1 0 1 2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0^{+20}_{-10}$
121 prop_binCROS_electron2018_bin19_Fake
122 prop_binCROS_electron2018_bin7_TTTo2L2Nu
123 prop_binCROS_electron2018_bin19_TTTo2L2Nu
124 prop_binCROS_electron2016M_bin21_Triboson
125 QCDScale_Triboson
126 prop_binCROS_electron2018_bin19_VG
127 QCDScale_TVX
128 prop_binCROS_electron2018_bin21_Triboson
129 prop_binCROS_electron2017_bin11_TTTo2L2Nu
130 prop_binCROS_electron2016M_bin9_WZ
131 prop_binCROS_electron2016M_bin7_WZ
132 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
133 prop_binCROS_electron2017_bin11_WZ
134 prop_binCROS_electron2017_bin9_TTTo2L2Nu
135 prop_binCROS_electron2018_bin7_TVX
136 prop_binCROS_electron2016M_bin19_Triboson
137 FR_sys_electron_2017
138 prop_binCROS_electron2017_bin17
139 prop_binCROS_electron2016M_bin20_Triboson
140 prop_binCROS_electron2017_bin9_TVX
141 prop_binCROS_electron2018_bin19_TVX
142 prop_binCROS_electron2016M_bin20_TVX
143 prop_binCROS_electron2018_bin21_TVX
144 prop_binCROS_electron2016M_bin8_ZZtoLep
145 prop_binCROS_electron2016M_bin20_ZZtoLep
146 prop_binCROS_electron2017_bin12_TTTo2L2Nu
147 prop_binCROS_electron2018_bin19_Triboson
148 prop_binCROS_electron2017_bin13_TTTo2L2Nu
149 prop_binCROS_electron2017_bin13_VG
150 QCDScale_VG
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0 +20 −10
151 prop_binCROS_electron2016M_bin11_TVX
152 prop_binCROS_electron2016M_bin8_VG
153 prop_binCROS_electron2016M_bin7_Triboson
154 prop_binCROS_electron2018_bin21_ZZtoLep
155 prop_binCROS_electron2016M_bin8_Triboson
156 prop_binCROS_electron2016M_bin19_ZZtoLep
157 prop_binCROS_electron2016M_bin8_TVX
158 prop_binCROS_electron2016M_bin19_TVX
159 prop_binCROS_electron2018_bin18_Triboson
160 prop_binCROS_electron2018_bin12_Triboson
161 prop_binCROS_electron2017_bin11_Triboson
162 prop_binCROS_electron2018_bin18_TVX
163 prop_binCROS_electron2018_bin12_TVX
164 prop_binCROS_electron2016M_bin21_TVX
165 prop_binCROS_electron2018_bin12_ZZtoLep
166 prop_binCROS_electron2017_bin13_ZZtoLep
167 prop_binCROS_electron2016M_bin21_ZZtoLep
168 QCDScale_ZZtoLep
169 prop_binCROS_electron2018_bin18_ZZtoLep
170 prop_binCROS_electron2017_bin13_Triboson
171 prop_binCROS_electron2016M_bin7_TVX
172 prop_binCROS_electron2017_bin12_Triboson
173 prop_binCROS_electron2017_bin9_ZZtoLep
174 prop_binCROS_electron2018_bin19_ZZtoLep
175 prop_binCROS_electron2017_bin11_TVX
176 prop_binCROS_electron2016M_bin7_ZZtoLep
177 prop_binCROS_electron2016M_bin11_Triboson
178 prop_binCROS_electron2017_bin12_TVX
179 prop_binCROS_electron2018_bin7_ZZtoLep
180 prop_binCROS_electron2017_bin12_ZZtoLep
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
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Poisson
AsymmetricGaussian
CMS Internal
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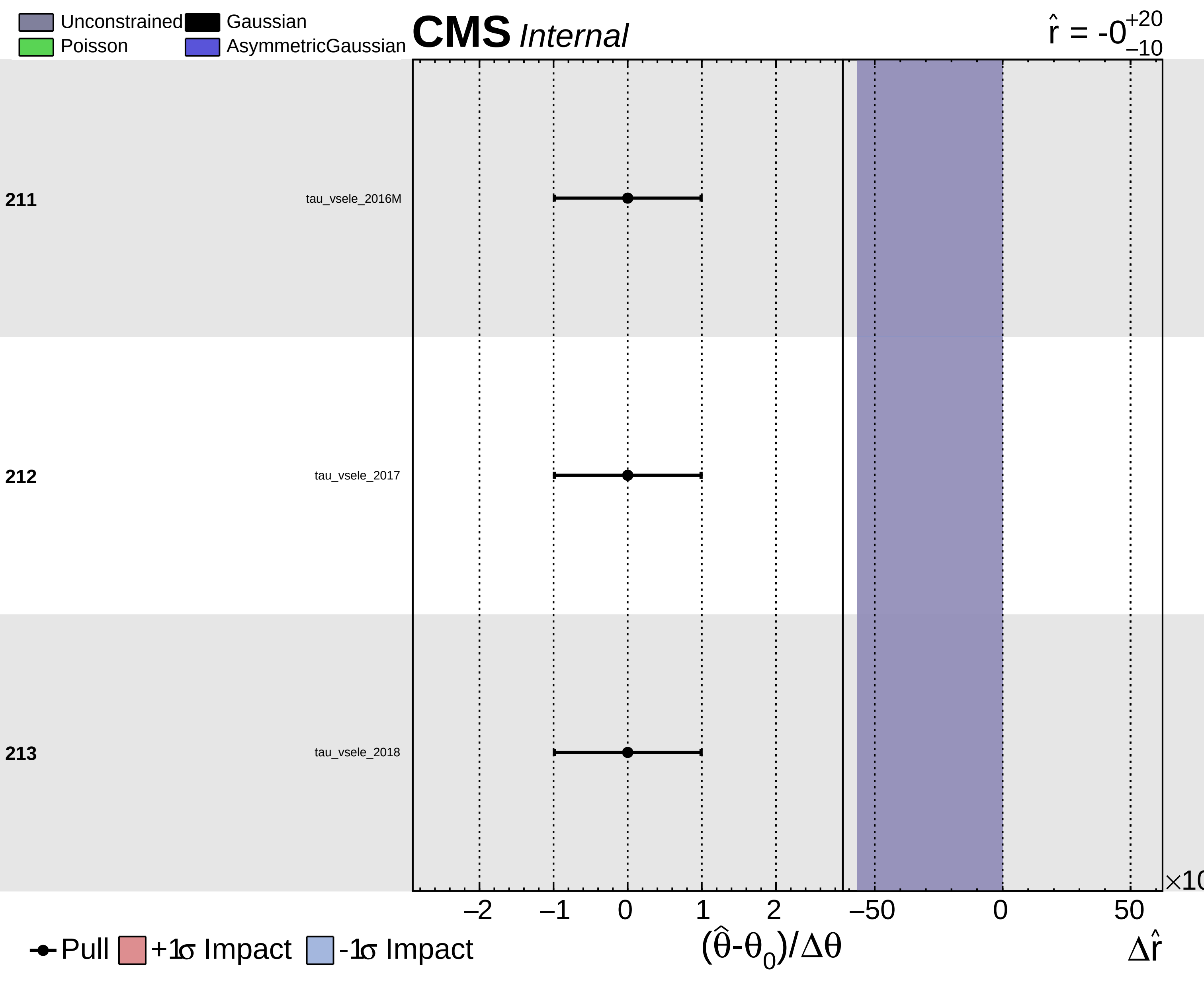
182 QCDScale_WpWpJJ_QCD
183 prop_binCROS_electron2017_bin9_Triboson
184 prop_binCROS_electron2016M_bin11_ZZtoLep
185 prop_binCROS_electron2017_bin11_ZZtoLep
186 prop_binCROS_electron2017_bin13_TVX
187 FES
188 prop_binCROS_electron2016M_bin9_ZZtoLep
189 prop_binCROS_electron2016M_bin9_TVX
190 prop_binCROS_electron2016M_bin9_Triboson
191 prop_binCROS_electron2017_bin12_WpWpJJ_EWK
192 prop_binCROS_electron2016M_bin11_WpWpJJ_EWK
193 QCDScale_sig
194 prop_binCROS_electron2017_bin13_WpWpJJ_EWK
195 VBS_2017
196 prop_binCROS_electron2016M_bin19_WpWpJJ_EWK
197 prop_binCROS_electron2016M_bin8_WpWpJJ_EWK
198 VBS_2016M
199 prop_binCROS_electron2017_bin9_WpWpJJ_EWK
200 prop_binCROS_electron2018_bin19_WpWpJJ_EWK
201 prop_binCROS_electron2018_bin7_WpWpJJ_EWK
202 VBS_2018
203 prop_binCROS_electron2018_bin12_WpWpJJ_EWK
204 prop_binCROS_electron2018_bin18_WpWpJJ_EWK
205 prop_binCROS_electron2017_bin11_WpWpJJ_EWK
206 prop_binCROS_electron2016M_bin20_WpWpJJ_EWK
207 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
208 prop_binCROS_electron2018_bin21_WpWpJJ_EWK
209 prop_binCROS_electron2016M_bin21_WpWpJJ_EWK
210 QCDScale_WZ
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0^{+20}_{-10}$
211
tau_vsele_2016M
212
tau_vsele_2017
213
tau_vsele_2018
−2
−1
0
1
2
−50
0
50
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$